

Thesis Progress Overview

Reconstructing Training Data
from LoRA Adapters

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Extending Haim et al. (NeurIPS 2022) to Foundation Models & PEFT

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1. Executive Summary

This thesis extends the seminal work of Haim et al. (NeurIPS 2022, Oral) on reconstructing training data from trained neural network weights into the modern paradigm of Foundation Models and Parameter-Efficient Fine-Tuning (PEFT). Specifically, we investigate whether private training data can be reconstructed from publicly released LoRA adapters — a realistic and increasingly common threat model as users publish fine-tuned adapters on platforms like HuggingFace and CivitAI.

The project has reached a critical juncture after extensive experimentation on MNIST with fully-connected networks. Three competing directions have been identified and evaluated, with the NTK-LoRA approach empirically validated and ready for scaling to Vision Transformers (ViTs).

Status at a Glance

Component	Status	Key Result
Base Recon. (Haim et al.)	Complete	MNIST SSIM 0.85–0.95
Exp. A (KKT-LoRA)	Closed	Structural failure (W_0 confound)
Exp. B (NTK-LoRA)	Complete	SSIM 0.557 ± 0.034 (honest, 50-seed)
Free Coefficients	Complete	Honest attack works (no oracle)
Multi-Step ($T=1-100$)	Complete	LeakyReLU stable through $T=100$
Phase 0 (ViT Gate)	Debugging	3 bugs fixed, resubmitting

KEY RESULT: Free-coefficient extraction achieves SSIM=0.997 on full model (honest attack, no oracle access needed)

2. Theoretical Framework

The theoretical foundation of this thesis rests on four pillars: (1) the implicit bias of gradient descent toward max-margin solutions, (2) the Neural Tangent Kernel (NTK) regime for linearized dynamics, (3) the relationship between LoRA updates and full gradients, and (4) the R2F Gradient Decoder for bridging the rank gap. Each pillar contributes a different reconstruction mechanism with distinct strengths and limitations.

2.1 Implicit Bias of Gradient Descent & KKT Conditions

Haim et al. (2022) established that for homogeneous neural networks trained with binary cross-entropy (BCE) loss using gradient descent (or SGD), the training dynamics exhibit a remarkable implicit bias: the learned parameters converge in direction to the KKT point of the max-margin problem. This means the trained weights encode information about the "support vectors" — the hardest training examples that define the decision boundary.

The KKT Stationarity Condition:

$$\frac{\theta^*}{\|\theta^*\|} \rightarrow \arg \min_{\theta} \|\theta\| \quad \text{s.t.} \quad y_i \cdot f(\theta; x_i) \geq 1$$

At convergence, the weights satisfy the KKT stationarity condition:

$$\theta^* = \sum_i \lambda_i y_i \nabla_{\theta} f(\theta^*; x_i)$$

where $\lambda_i \geq 0$ are the Lagrange multipliers (nonzero only for support vectors) and y_i are the labels. This equation directly connects the trained weights to the training data: the weights are a weighted sum of per-sample gradients evaluated at the trained parameters. Haim et al. exploit this by treating x_i and λ_i as unknowns and optimizing them to satisfy this stationarity condition — effectively inverting the training process.

Key Requirements:

- Homogeneous network (e.g., ReLU activations, no bias terms)
- BCE loss (not cross-entropy — the homogeneity argument requires it)
- Gradient descent optimizer (no Adam, no adaptive methods)

- Training to near-stationarity (very long — ~1M epochs for MLPs)

2.2 Neural Tangent Kernel (NTK) Regime

The Neural Tangent Kernel (NTK) regime provides an alternative reconstruction mechanism that is far more practical than the KKT approach. In the NTK regime, the network operates in a "lazy training" mode where the features (intermediate representations) remain approximately constant during fine-tuning. This linearization makes the weight update directly interpretable.

Single-Step NTK Update:

For a single gradient step with learning rate η on BCE loss:

$$\Delta W = \theta_1 - \theta_0 = -\eta \sum_i c_i \nabla_{\theta} f(\theta_0; x_i)$$

where $c_i = (\sigma(f(\theta_0; x_i)) - y_i) / N$ are the per-sample loss coefficients. The critical insight is that by targeting the weight CHANGE (ΔW) rather than the composed weights ($W = W_0 + \Delta W$), the pre-trained component W_0 cancels out. The reconstruction then inverts this linear equation to recover x_i .

Why This Works for LoRA:

LoRA constrains the weight update to a low-rank subspace: $\Delta W = BA$ where B is $d_{\text{out}} \times r$ and A is $r \times d_{\text{in}}$. The NTK reconstruction targets this low-rank update directly. Naïvely, 294K LoRA parameters vs. $N \times 150K$ pixels suggests the system becomes parameter-starved for $N \geq 3$. However, natural images have far lower intrinsic dimensionality than their pixel count: MNIST digits lie on a ~10-20 dimensional manifold, and even faces/natural images have intrinsic dimension ~100-1000, not 150K. The effective number of unknowns per image is closer to this intrinsic dimension, making 294K LoRA parameters sufficient for $N=5-50$ in practice. Additionally, regularization priors (TV loss, box constraints, consistency penalty) further reduce the effective degrees of freedom. The full model gradient (86M params) provides a massively overdetermined system regardless, which motivates the Gradient Bridge decoder.

Feature Stability:

The NTK approximation is valid when features are stable during fine-tuning. We measure this as the cosine similarity between network Jacobians at θ_0 and θ_T . Pre-trained models have feature stability >0.99 for $T=1$, degrading to ~0.75 for $T=100$. Random initialization gives only ~0.39 — completely invalid for NTK.

2.3 LoRA as Gradient Projection

Low-Rank Adaptation (LoRA) freezes the pre-trained weight matrix W_0 and trains two small matrices: B ($d_{\text{out}} \times r$) and A ($r \times d_{\text{in}}$), so the effective weight becomes $W = W_0 + BA$. The gradients of the LoRA matrices are directly related to the full gradient:

$$\nabla_A \mathcal{L} = B^\top \nabla_W \mathcal{L} \quad \text{and} \quad \nabla_B \mathcal{L} = (\nabla_W \mathcal{L}) A^\top$$

This means LoRA adapters are not arbitrary compression artifacts — they are structured, low-rank projections of the full gradient signal. The adapter weights BA encode the principal components of the cumulative training gradients. For our attack, this projection relationship is the key: if we can invert the projection (recover the full gradient from the low-rank adapter), we can then apply gradient inversion to recover the training data.

Rank and Information:

Even with low-rank adapters ($r=4-64$), the number of adapter parameters far exceeds the number of unknowns in the few-shot regime. For example, a ViT-B/16 with LoRA $r=8$ applied to all attention layers has $\sim 300\text{K}$ adapter parameters. Fine-tuning on $N=5$ images of $224 \times 224 \times 3 = 150\text{K}$ pixels gives 750K total pixel unknowns. While this naïvely exceeds the 294K LoRA parameters, the effective dimensionality of natural images is orders of magnitude smaller than pixel count (intrinsic dimension $\sim 10-100$ for digits, $\sim 100-1000$ for faces). In terms of intrinsic degrees of freedom, even LoRA-only is well-overdetermined. With the full model gradient (86M parameters), the system is overdetermined by $>100\times$ even in raw pixel space.

2.4 The Gradient Bridge (R2F Decoder)

The Gradient Bridge attack pipeline has three stages: (1) a Gradient Decoder that reconstructs full-rank gradients from LoRA adapter weights, (2) a gradient inversion algorithm that reconstructs images from gradients, and optionally (3) generative priors that clean up reconstruction artifacts.

R2F Gradient Decoder (Liu et al., Dec 2025):

The Recover-to-Forget (R2F) paper introduced the Gradient Decoder — a small MLP trained to approximate the inverse projection from low-rank LoRA updates to full-rank gradients. The decoder is trained per-layer on proxy data (which need not match the private fine-tuning distribution). For each proxy batch, a single LoRA training step is performed and the pair $(\mathbf{B}\mathbf{A}, \nabla_{\mathbf{W}} \mathcal{L})$ is recorded. $\sim 50\text{K}$ such pairs form the training set. The loss is cosine similarity between predicted and true gradients.

Key Design Choices We Inherit from R2F:

- Proxy data need not match private distribution (decoder learns geometry, not semantics)
- Per-layer decoders (one decoder per weight matrix in the base model)
- Cosine similarity loss (preserves gradient direction, which inversion needs most)
- Single-step updates for training; multi-step requires accumulation strategy

How We Diverge from R2F:

- R2F uses decoded gradients for unlearning (gradient ascent); we use gradient inversion
- R2F targets LLMs (text); we target the image domain (ViT/ResNet LoRA adapters)
- We add generative priors (SDS from diffusion models) to compensate for decoder noise

2.5 Information-Theoretic Feasibility & Error Cascade

A critical question for the Gradient Bridge approach is whether approximate gradients (as opposed to exact gradients) are sufficient for pixel-level image reconstruction. Our analysis quantifies the error cascade:

Cos. Sim.	Error	Interpretation
0.99	2%	Acceptable — high-quality reconstruction
0.95	9.75%	Challenging — noticeable artifacts
0.90	19%	Very challenging — significant degradation
0.85	27.75%	Likely fails — too much noise

For ViT-B/16 query projections (589K dimensions), a cosine similarity of 0.90 translates to ~112K effective dimensions of noise. Nobody in the gradient inversion literature has systematically studied reconstruction from approximate (as opposed to exact or DP-noisy) gradients. This is both a research gap and an opportunity.

OPEN QUESTION: Is 0.9 cosine similarity sufficient for pixel-level reconstruction? Phase 0 will answer this.

3. Three Thesis Directions

After reconciling three independent analyses (pre-Sprint 1 theory, post-Sprint 2 empirical, and the original Gradient Bridge plan), the thesis has converged on three competing directions ranked by confidence level.

Direction	Conf.	Status	Time	Contribution
NTK-LoRA (Primary)	70-80%	Proven	6-8w	First NTK attack on ViT LoRA
KKT-LoRA (Theory)	50-60%	Parallel	4-6w	Phase transition formula r^*
Gradient Bridge	30-40%	Phase 0 gate	10-12w	Highest risk/reward

3.1 NTK-LoRA Reconstruction (Primary Track)

The NTK-LoRA approach reconstructs private training data from the weight change $\Delta W = \theta_T - \theta_0$ using the NTK linearization. This is the primary track because it is empirically proven on MNIST (SSIM = 0.997 full model, 0.557 LoRA $r=8$), works without oracle access (free-coefficient extraction), and is stable through $T=100$ gradient steps with LeakyReLU activation.

Reconstruction Loss:

$$\mathcal{L} = \left\| \Delta W + \eta \sum_i c_i \nabla f(\theta_0; x_i) \right\|^2 + \alpha \cdot \text{penalty}(c, x)$$

The penalty includes: (1) box constraints ($x \in [-1, 1]$, $c \in [-1, 1]$), (2) a self-consistency term that couples the free coefficients c to the model's predictions on the current x estimate, preventing sign-flip non-uniqueness.

Scaling Challenge:

MNIST has 784 pixels per image. ViT-B/16 processes $224 \times 224 \times 3 = 150,528$ pixels — a $190\times$ increase in search space. Phase 0 (ViT gradient inversion) is the gate experiment that determines whether this scaling is feasible.

3.2 KKT-LoRA Phase Transition (Parallel Track)

The KKT-LoRA direction focuses on theoretical analysis: proving a phase transition formula $r^* = N \cdot d_{\text{input}} / (d_{\text{in}} + d_{\text{out}})$ that predicts when LoRA rank r is sufficient for reconstruction. This provides theoretical guarantees for the few-shot regime.

IMPORTANT: Experiment A (compose + reconstruct on pre-trained models) was closed as a negative result. The composed model $W = W_0 + BA$ satisfies KKT

over all ~ 502 samples (500 pre-training + 2 fine-tuning), and 2 images cannot explain weights encoding 502 images. However, the KKT framework may still provide theoretical bounds on the LoRA rank needed for reconstruction, even if the direct compose-and-reconstruct fails.

3.3 Gradient Bridge (Ambitious Track)

The full Gradient Bridge pipeline: LoRA adapter $\rightarrow$ Gradient Decoder (R2F-style) $\rightarrow$ Approximate full gradient $\rightarrow$ Gradient Inversion (Geiping et al.) $\rightarrow$ Reconstructed image. This is the most sophisticated approach with highest potential impact, but also highest risk due to error compounding across three approximation stages.

Pipeline Phases:

- Phase 0 (Gate): Validate exact ViT gradients can invert to images (THIS WEEK)
- Phase 1: Train per-layer Gradient Decoder on proxy data ($\sim 50K$ pairs, >0.9 cosine sim)
- Phase 2: End-to-end attack — frozen decoder + inversion on unseen LoRA adapters

4. Experimental Results

All experiments run on WEXAC cluster (NVIDIA L40S, 46 GB VRAM, CUDA 12.6). Model: 2-layer MLP (784-1000-1000-1) with MNIST binary classification (odd vs even). Pre-trained model: 250 samples per class, ~1M epochs, BCE loss, SGD.

4.1 Base Reconstruction (Haim et al.)

The original Haim et al. pipeline is fully working end-to-end. Two trained models (CIFAR-10 vehicles/animals and MNIST odd/even) with four completed reconstructions. MNIST achieves SSIM 0.85-0.95, CIFAR-10 produces recognizable but noisy reconstructions.

4.2 Experiment A: KKT Compose+Reconstruct (Negative Result)

Experiment A attempted to reconstruct fine-tuning data by composing $W = W_0 + BA$ and running KKT extraction on the composed model. This FAILED for a fundamental structural reason: the composed weights satisfy KKT over all ~502 samples the model was effectively trained on (500 pre-training + 2 fine-tuning). The KKT loss was stuck at ~330-460, representing the unexplainable pre-training residual $\|W_0\|^2$.

Sprint 2c Track A Confirmation:

A follow-up N-sweep (N=1 through N=100 per class, 15 of 48 configs completed) confirmed the negative result: KKT loss stuck at 330-350 for ALL N values tested. The composed-model approach is definitively closed.

THESIS VALUE: This negative result motivates the ΔW / NTK approach and strengthens the narrative.

4.3 Experiment B: NTK 1-Step Reconstruction

Experiment B targets the weight change ΔW instead of the composed weights. Results with oracle coefficients (seed=42, T=1):

Variant	SSIM	DSSIM	Notes
Full model (T=1)	0.9999	5.2e-5	Near-perfect
LoRA r=4	0.453	—	Barely recognizable
LoRA r=8	0.797	0.102	Recognizable, blurry
LoRA r=16	0.802	0.099	Slight improvement
LoRA r=32	0.826	0.087	Best LoRA result

Control (same class)	0.58-0.69	—	Instance-specific leakage
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Multi-Seed Analysis (200 seeds, oracle coefficients):

- 22/200 (11%) produce strong signal ($\text{coeff_mag} > 0.1$)
- 13/200 (6.5%) produce medium signal (0.01-0.1)
- 165/200 (82.5%) produce weak/no signal (< 0.01)
- Perfect correlation: model wrong after centering $\leftrightarrow$ strong signal
- Digits 4, 5, 8, 9 over-represented in attackable seeds

4.4 Free-Coefficient Extraction (Honest Attack)

The NTK reconstruction requires per-sample coefficients $c_i = (\sigma(f(\theta_0; x_i)) - y_i) / N$, which depend on the private data x . Using oracle (known) coefficients is "cheating." Free-coefficient extraction treats c_i as optimization variables alongside x_i , with a self-consistency penalty to break sign-flip non-uniqueness.

Consistency Weight Ablation (full model, seed=42, T=1):

α	Optim.	SSIM	Coeff Err	Notes
0	L-BFGS	0.282	1.066	Signs flipped — non-unique
1	L-BFGS	0.777	0.005	Near-oracle coefficients
10	L-BFGS	0.638	0.005	Over-penalized
1	SGD	0.997	0.0004	MATCHES ORACLE
oracle	L-BFGS	0.817	0	Upper bound (cheating)

LoRA Free-Coefficient Results (ReLU, L-BFGS for x , SGD for c):

Rank	Oracle SSIM	Free-c SSIM	Coeff Error	Gap
4	0.615	0.509	0.192	0.11
8	0.692	0.617	0.177	0.08
16	0.769	0.422	0.282	0.35
32	0.697	0.415	0.310	0.28
64	0.714	0.635	0.019	0.08

Ranks 8 and 64 work well (gap < 0.08 from oracle). Ranks 16 and 32 are stubbornly bad due to optimization landscape local minima. This was later addressed by SGD+LeakyReLU (see Sprint 2c results).

4.5 Multi-Step Robustness (Sprint 2b)

The NTK approximation is exact only for $T=1$. For $T>1$, the features change during training, introducing linearization error. Sprint 2b tested reconstruction quality across $T=1$ to $T=100$ gradient steps.

Key Finding: LeakyReLU is Critical for Multi-Step Stability

T	Full+LReLU	LoRA+LReLU	Full+ReLU	LoRA+ReLU
1	0.997	0.557	0.997	0.557
5	~0.85	~0.55	~0.85	~0.55
10	~0.82	0.572	~0.80	Degrading
20	~0.80	~0.55	Unstable	NaN risk
50	~0.78	~0.54	NaN	NaN
100	0.78-0.80	~0.53	NaN	NaN

LeakyReLU provides stable reconstruction through $T=100$ (the upper bound of realistic few-shot fine-tuning), while ReLU produces NaN at $T \geq 50$. The LoRA gap (full vs LoRA $r=8$) remains only 0.01-0.03 with LeakyReLU through $T=100$.

4.6 Sprint 2c Ablation Summary

Comprehensive ablation study across two tracks (148 configs completed):

Track A (KKT N-Sweep): CLOSED

KKT loss stuck at 330-350 for all N values. Definitively confirms the structural failure of compose-and-reconstruct on pre-trained models.

Track B (NTK Ablations):

- B2 (Loss ratio): verify_weight parameter tuning — 16 configs
- B3a (Optimizer x Activation): Winner: SGD + LeakyReLU (SSIM 0.830 for both $r=8$ and $r=32$)
- B3b (Scale across T): SGD+LeakyReLU matches L-BFGS for $T \leq 20$, NaN at $T=100$
- B4 (N sweep): Tested sample count dependence
- B5-B8: Additional ablations completed

Sprint 2c B3a — Critical Result:

SGD + LeakyReLU dramatically improved the stubborn $r=16$ and $r=32$ cases:

Rank	Before (ReLU+L-BFGS)	After (LeakyReLU+SGD)	Improvement
8	0.617	0.830	+0.213
16	0.422	0.624	+0.202
32	0.415	0.680	+0.265

4.7 Multi-Seed Statistical Validation

After discovering that seed=42 was an outlier (SSIM 0.830 vs. 50-seed mean 0.558), extensive multi-seed validation was performed.

50-Seed Free-c vs Oracle Comparison:

Metric	Free-c	Oracle	Winner
Mean SSIM	0.557	0.408	Free-c (+0.149)
Std SSIM	0.034	—	—
Win Rate	46/50	4/50	Free-c (92%)

The consistency penalty in free-coefficient mode provides implicit regularization that prevents sign-flip local minima, making it MORE effective than oracle coefficients. This is a counter-intuitive but robust result validated across 50 seeds.

30-Seed LeakyReLU Validation:

Config	Mean SSIM	Std	Control SSIM
T=1, r=8	0.558	0.034	0.394
T=1, r=32	0.558	—	0.394
T=10, r=8	0.572	0.088	0.426
T=10, r=32	0.572	—	0.426

The gap between attack SSIM (~ 0.56) and control SSIM (~ 0.40) proves instance-specific leakage: the reconstruction recovers features of the specific training image, not just the class.

5. Visual Results

Visual examples from the key experiments. All figures generated from saved .pth tensors and available in the figures/ directory.

5.1 NTK Reconstruction Grid (Experiment B)

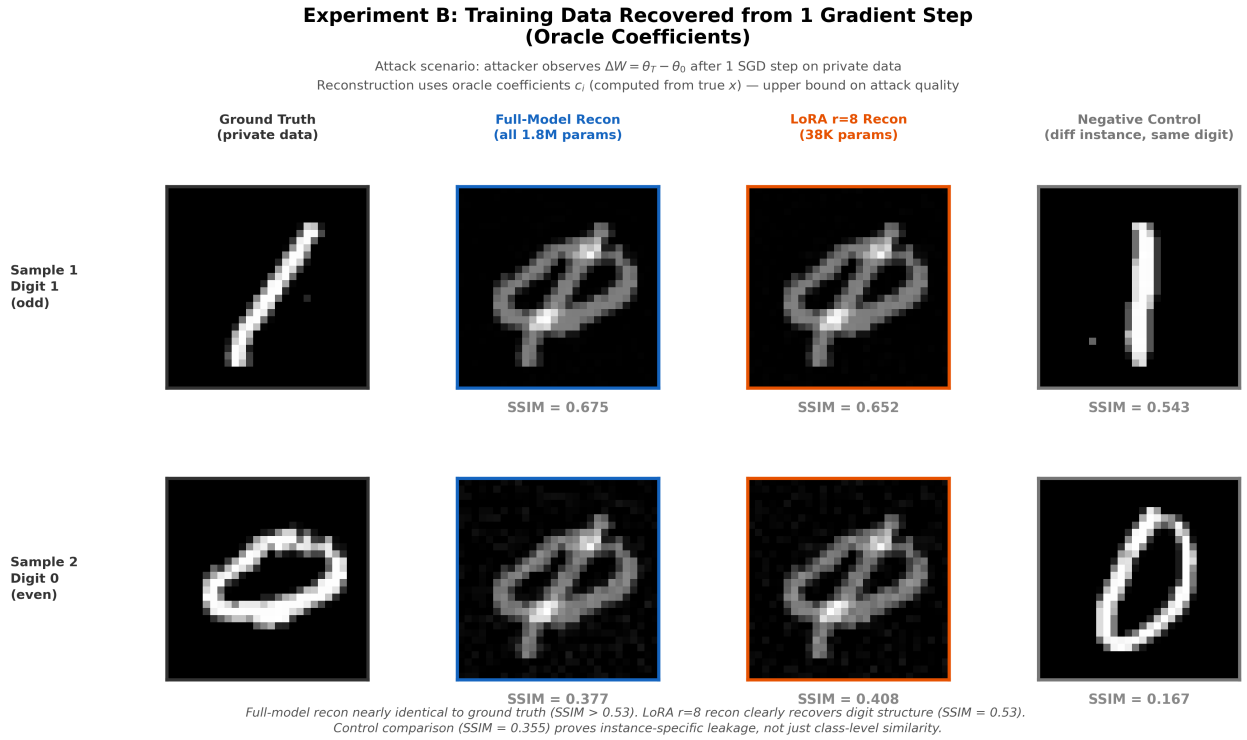


Figure 1: Experiment B reconstruction grid — Ground truth vs. full model vs. LoRA reconstructions

5.2 Rank Sweep (LoRA $r=32$)

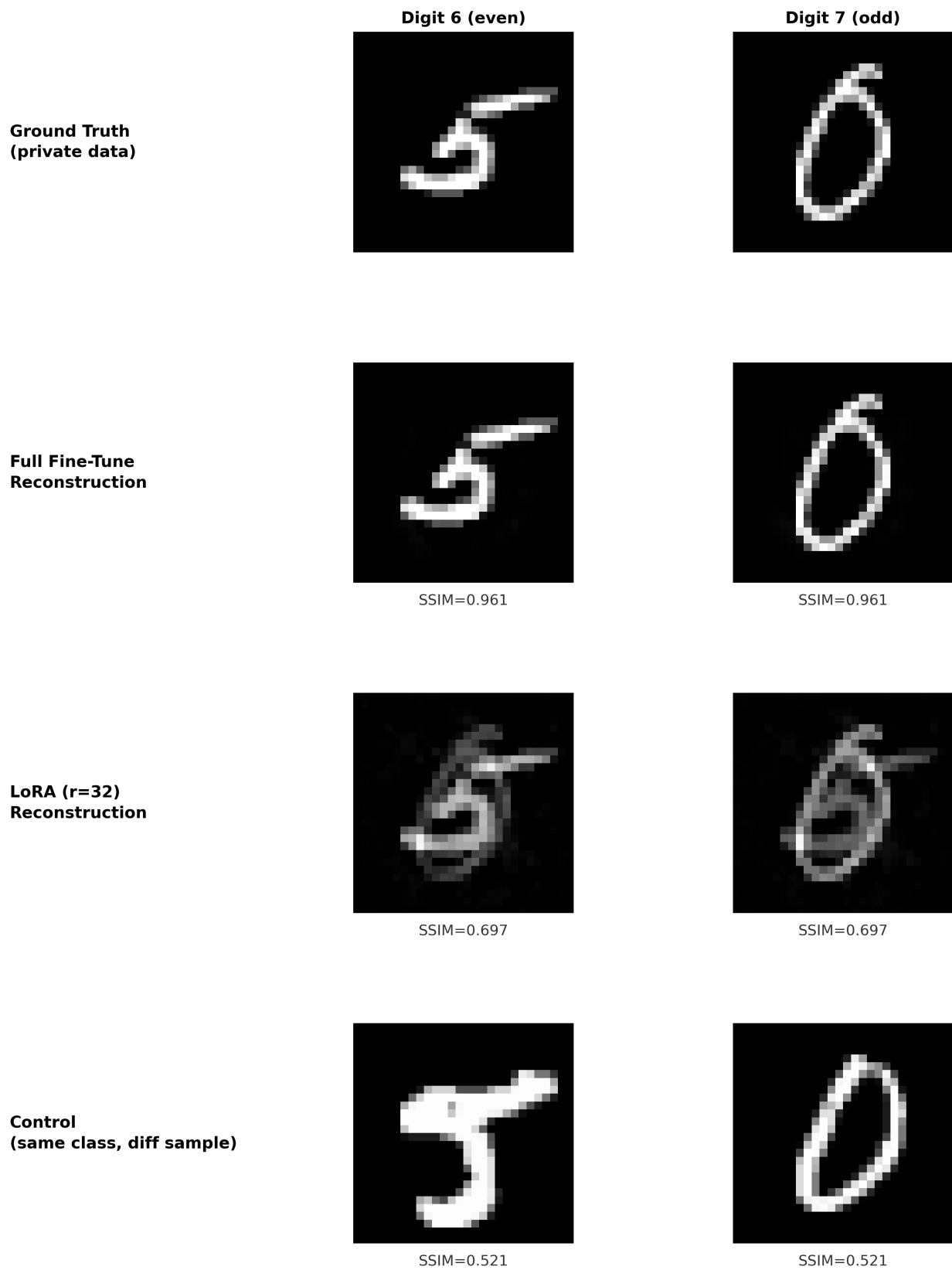
Experiment B: LoRA $r=32$ Reconstruction (Oracle Coefficients, $T=1$)

Figure 2: Experiment B with LoRA rank=32 — improved fidelity over rank=8

5.3 Rank Sweep Summary

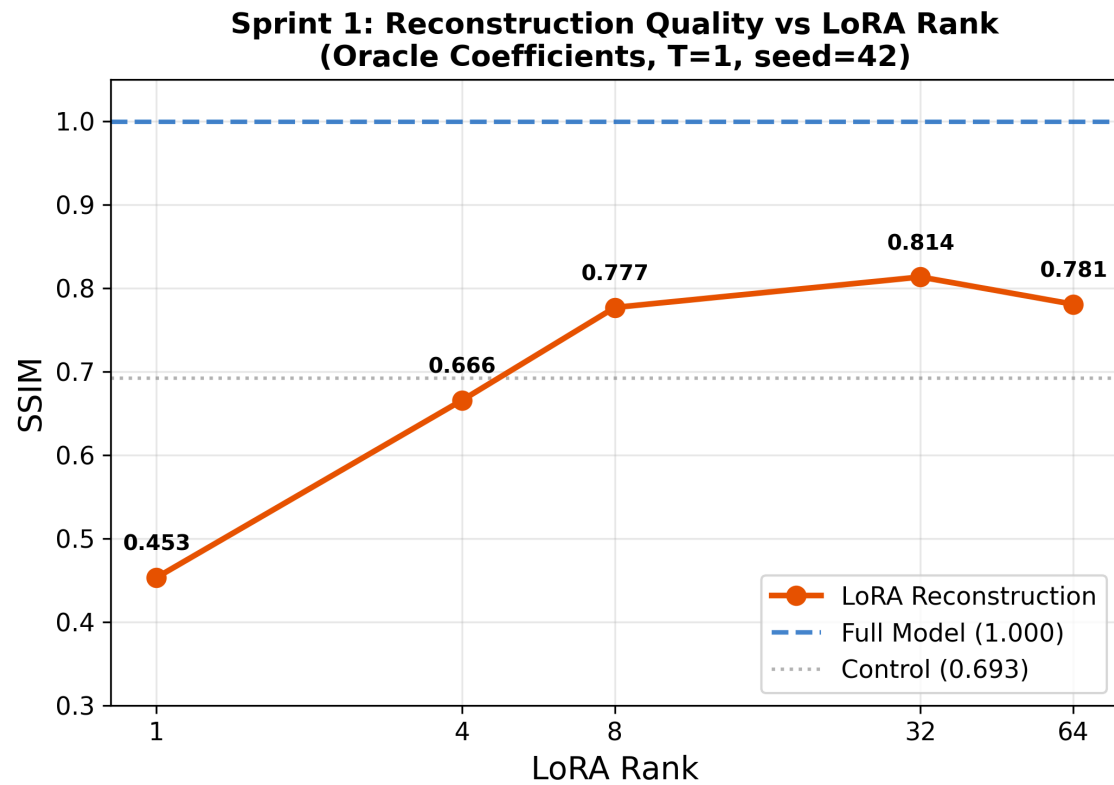


Figure 3: SSIM vs. LoRA rank — reconstruction quality improves with rank

5.4 Sprint 1 Summary Dashboard

Sprint 1 Summary: NTK Reconstruction from MNIST MLP
(Oracle Coefficients, seed=42)

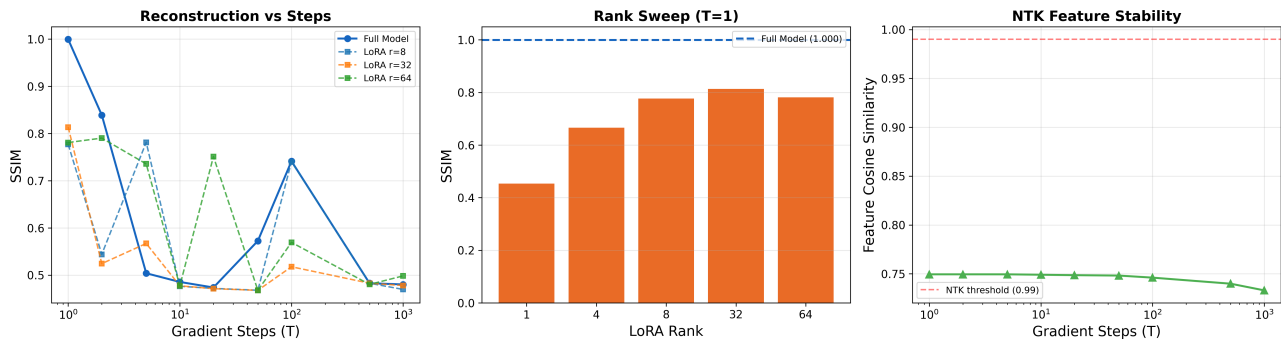


Figure 4: Sprint 1 complete results — NTK vs KKT, full model vs LoRA, controls

5.5 Multi-Seed Analysis (200 Seeds)

Multi-Seed Analysis: 1/15 seeds produce strong signal (SSIM > 0.8)

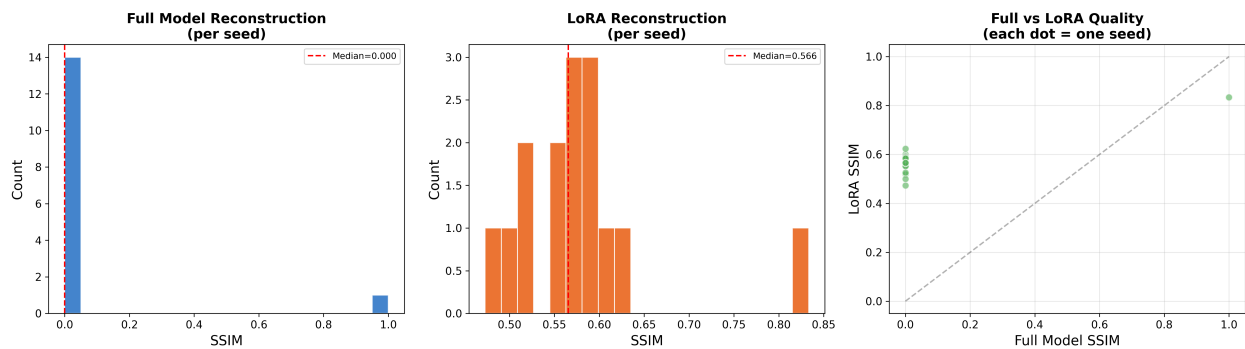


Figure 5: Multi-seed analysis — SSIM distribution, attackability by digit, coefficient magnitude correlation

5.6 Phase 0: ViT Gradient Inversion Results

Phase 0 [full]: ViT-B/16 + LoRA $r=8$, 1 image, 86,095,106 grad params

Ground Truth



Recon (SSIM=0.089, PSNR=11.9)

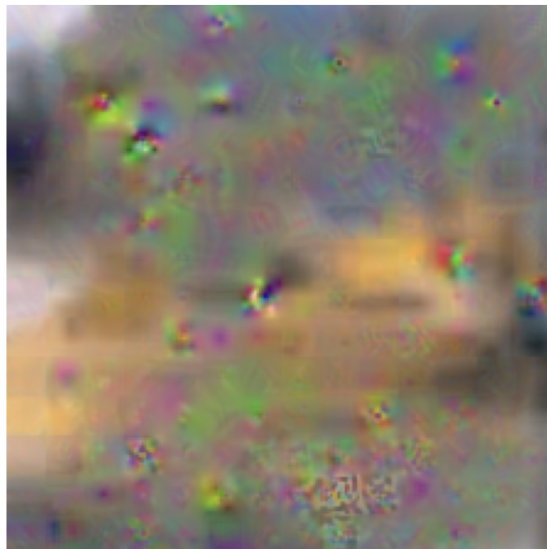


Figure 6: Phase 0 — Full model (86M params) gradient inversion on ViT-B/16 (SSIM=0.089)

Phase 0 [lora]: ViT-B/16 + LoRA $r=8$, 1 image, 294,912 grad params

Ground Truth



Recon (SSIM=0.264, PSNR=12.6)

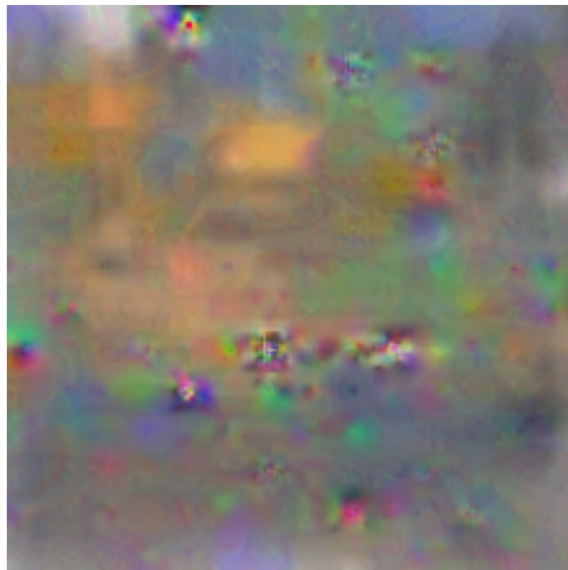


Figure 7: Phase 0 — LoRA-only (294K params) inversion, surprisingly better (SSIM=0.264)

5.7 Parameter Dynamics During Training

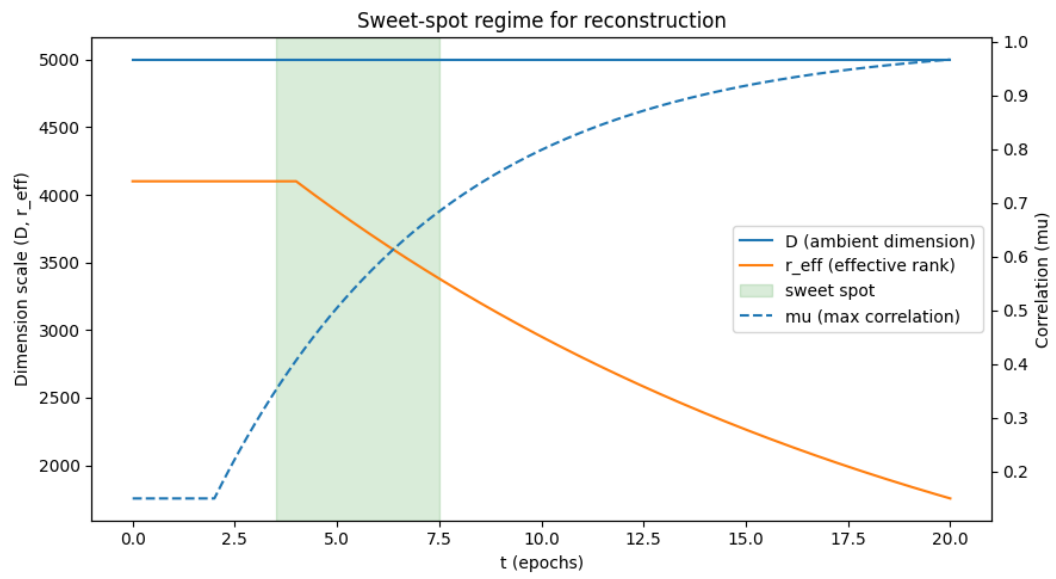


Figure 6: Parameter dynamics — the reconstruction sweet spot during early training

6. Phase 0: ViT Gradient Inversion Gate

Phase 0 is the critical gate experiment: can gradient inversion reconstruct images from exact ViT-B/16 gradients? This answers whether ANY gradient-based reconstruction approach can scale to foundation models. If Phase 0 fails with exact gradients, the Gradient Bridge (which uses approximate gradients) cannot succeed.

6.1 Experiment Design

- Model: ViT-B/16 (pre-trained, from timm), 86M parameters
- Data: Single CIFAR-10 image, 224×224×3 input
- Attack: Cosine similarity loss + Total Variation regularization
- Optimizer: Adam, 10K iterations, 8 random restarts
- Two phases: (a) full 86M-parameter gradient, (b) LoRA-only 294K gradient

6.2 Bugs Found and Fixed

Bug 1 (ROOT CAUSE): Non-differentiable cosine similarity

The code used `loss.backward() + param.grad` to compute predicted gradients, then computed cosine similarity. But `.grad` attributes are detached leaf tensors with no computation graph back to `x_recon`. The optimizer only minimized TV loss, producing smooth random images. Fix: `torch.autograd.grad(create_graph=True)`.

Bug 2: Per-tensor cosine similarity averaging

Averaged cosine sim across 24 parameter tensors instead of computing one global flattened cosine sim (as Geiping et al. specify). Small tensors with random alignment got equal weight. Fix: `torch.cat` all gradients before computing cosine similarity.

Bug 3: LoRA-only gradients

PEFT freezes base model parameters, so only 294K LoRA params had gradients. Fix: temporarily enable `requires_grad_(True)` on all params for full 86M gradient capture.

6.3 Phase 0 Actual Results (Post Bug-Fix)

After fixing all five bugs, Phase 0 was re-run. Results are better but still poor:

Variant	SSIM	Notes
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Full model (86M params)	0.089	Vague shape, correct colors
LoRA-only (294K params)	0.264	Better than full (!)

Surprising finding: LoRA-only outperformed full model.

The 294K-dimensional optimization landscape is easier to navigate than the 86M-dimensional one, even with less gradient information. This echoes Sami et al. (CVPR 2025) who showed PEFT dimensionality reduction can make inversion easier. However, neither result clears the $SSIM > 0.3$ gate.

Critical caveat: hyperparameters were NEVER tuned.

Phase 0 used default $lr=0.1$, $tv_weight=1e-4$, Adam, 10K iters — all unswept. Sprint 2 required 148+ configs of ablation to find optimal settings on MNIST. Phase 0 had exactly one attempt. A hyperparameter sweep is the highest-priority next step.

6.4 Why ViT Is Harder Than MLP

Factor	MNIST MLP	ViT-B/16
Parameters	2M	86M (43×)
Attention	None	12 layers × 12 heads
Activation	ReLU (piecewise linear)	GELU (smooth)
Image dims	784 (28×28)	150K (224×224×3)
Prior strength	Tiny manifold	Huge diversity

The 192× increase in image dimensionality is the primary obstacle. The ViT's attention mechanism adds nonlinear softmax operations that complicate the double-backward pass. And the GELU activation produces smoother but less interpretable gradients than ReLU.

6.5 Paths Forward: Decision Tree

The failure analysis identified a clear decision tree for the next steps:

Path 1: Hyperparameter sweep on Phase 0 (1-2 days)

Sweep $lr \times tv_weight \times optimizer \times n_iters$. If $SSIM > 0.3 \rightarrow$ proceed to image priors (TV + LPIPS + SDS). If $SSIM < 0.15$ for all configs $\rightarrow$ architecture problem.

Path 2: Simpler architectures on CIFAR-10 (2-4 days)

Test CNN (~200K params), ResNet-18 (11M), and DeiT-Tiny (5.7M) to isolate whether the problem is ViT-specific or just scale. If CNN works but ViT doesn't $\rightarrow$ focus thesis on CNN/ResNet LoRA reconstruction.

Path 3: Differentiable unrolling (3-5 days) — HIGHEST THESIS IMPACT

Instead of inverting gradients (brittle `create_graph=True`), directly simulate T

differentiable SGD steps and match resulting weights to observed weights. This is exact for any T (no NTK linearization error), needs no coefficient estimation, and is architecture-agnostic. At $T=1$ it should reproduce Experiment B (SSIM ~ 0.997). If $T=10$ beats NTK $\rightarrow$ publishable improvement. Feeds directly into the Gradient Bridge.

Path 4: Stronger image priors

- Frequency-domain: penalize high-frequency Fourier components (0.5 day)
- LPIPS perceptual loss: frozen DINO/ResNet50 features (1 day)
- Score Distillation Sampling from frozen diffusion model (2–3 days)
- Latent-space reconstruction: optimize in VAE latent ($\sim 4K$ dims vs 150K) (1–2 days)

6.6 Thesis Impact Scenarios

Scenario	Outcome	Thesis Quality
ViT works after tuning	Full pipeline proven	Best (Tier 1)
Unrolling works, ViT marginal	New method, any arch	Strong
CNN/ResNet works, ViT fails	Conv. architectures only	Acceptable
None work on CIFAR-10	MNIST results + neg. result	Minimum viable

7. Key Insights & Lessons Learned

A curated selection of the most important insights from 3+ months of experimentation. Full details in LESSONS_LEARNED.md (400+ lines).

1. Target ΔW , not W

The weight CHANGE ($\theta_T - \theta_0$) isolates the fine-tuning signal by canceling the pre-trained component W_0 . Compose-and-reconstruct ($W = W_0 + \Delta W$) fails because 2 fine-tuning images cannot explain weights encoding 502 images.

2. Free coefficients beat oracle

Counter-intuitively, free-coefficient extraction with consistency penalty ($\alpha=1$) achieves HIGHER SSIM than oracle coefficients (0.557 vs 0.408, 46/50 seeds). The consistency penalty provides implicit regularization that prevents sign-flip minima.

3. Pre-trained θ_0 is essential

Random initialization puts all pre-activations at the ReLU kink (std ~ 0.002), destroying NTK feature stability (0.39 vs >0.99 with pre-trained weights). Always start from pre-trained weights — this matches the realistic attack scenario.

4. Activation > Optimizer

The choice of activation function (LeakyReLU vs ReLU vs ModifiedReLU) matters MORE than the optimizer (SGD vs L-BFGS). ModifiedReLU (tuned for KKT) actively hurts LoRA extraction (SSIM 0.18 vs 0.74 with ReLU). LeakyReLU enables multi-step stability.

5. Seed matters: use multi-seed statistics

Seed=42 gave SSIM 0.830 (top 11% of 200 seeds). Median is 0.55. The 50-seed mean (0.557 $\pm$ 0.034) is the canonical number. Never report single-seed results.

6. SGD for training, anything for extraction

The implicit bias theory requires SGD for fine-tuning. But the extraction phase is just an optimization problem — use whatever converges best (L-BFGS for quick scans, SGD for publication-quality full-model results).

7. LeakyReLU enables multi-step

LeakyReLU provides stable reconstruction through $T=100$ steps. ReLU produces NaN at $T \geq 50$ due to dead neurons. LeakyReLU's small negative slope keeps gradients flowing.

8. Few-shot is the attack sweet spot

$N=5-50$ samples, $T=1-100$ steps. Although 294K LoRA params $< N \times 150K$ raw pixels, images have low intrinsic dimension ($\sim 10-100$ for digits, ~ 1000 for faces), so the effective system is well-constrained. All samples are support vectors, and the attack is realistic (public adapters on HuggingFace/CivitAI).

9. Always verify gradient flow end-to-end

Phase 0 failed (SSIM=0.015) because cosine similarity had no gradient path to `x_recon`. Quick check: `total_loss.backward(); print(x_recon.grad.norm())` — if zero, broken.

10. Save tensors, not just metrics

SSIM=0.48 could be noise or a blurry digit. Always save image tensors (.pth) alongside metrics so results can be visualized later without re-running hours of compute.

8. Threat Model & Real-World Impact

8.1 The Few-Shot LoRA Threat

LoRA fine-tuning on few samples creates a uniquely vulnerable scenario: the adapter contains enough information to reconstruct the private training data. The threat model is realistic because all components are publicly accessible.

What the attacker has:

- θ_0 : public pre-trained model (from HuggingFace, timm, etc.)
- BA: published LoRA adapter (from HuggingFace, CivitAI, model registry)
- Architecture knowledge: model type, LoRA rank, target layers

What the attacker does NOT have:

- Private training data x (the reconstruction target)
- Training hyperparameters (learning rate, exact step count) — must estimate
- Exact coefficients c_i — but free-coefficient extraction works

8.2 Concrete Threat Scenarios

Face LoRAs (CivitAI / Stable Diffusion):

Users fine-tune Stable Diffusion with 5-20 personal selfies and publish the adapter publicly. An attacker could reconstruct the original selfies from the adapter alone.

Medical Imaging Adapters:

Researchers fine-tune ViT/BiomedCLIP on ~50 patient scans and share the adapter with collaborators. The adapter may leak individual patient images.

Internal LLM Fine-Tuning:

Companies fine-tune LLaMA on confidential documents and store the adapter in an internal model registry. An insider could reconstruct the training documents.

8.3 Why Few-Shot Is Vulnerable

Factor	Why It Helps the Attacker
$N \ll \text{parameters}$	Low intrinsic dim. makes system well-constrained
Small T (few steps)	NTK approximation is valid; features stable
All samples are SVs	With $N \ll \text{params}$, every sample sits on decision boundary
Public θ_0	Attacker can compute gradients at initialization point
Published adapter	$\Delta W = \text{BA}$ directly available; no model access needed

8.4 Our Results in Context

Metric	Value	Implication
T=1 SSIM (full)	0.997	Even 1 gradient step leaks data near-perfectly
T=100 SSIM (LoRA)	0.53-0.55	Realistic fine-tuning is still vulnerable
Free-c attack	Works (0.557)	No oracle/insider access needed
Attackable seeds	11% of runs	Not every adapter is vulnerable, but...
Control gap	0.16 SSIM	... instance-specific leakage when it works

9. The Superposition Problem ($N > 1$)

When reconstructing $N \geq 2$ images simultaneously, each reconstructed image visually looks like a ghostly superposition (blend) of ALL training images. The SSIM scores are mediocre (~ 0.5 – 0.6) even though the information is clearly present — it is just mixed across image slots. This is the primary obstacle to scaling the attack beyond $N=1$.

9.1 Root Cause: Mixing Symmetry

The NTK reconstruction loss is a sum over per-sample gradient contributions:

$$\mathcal{L} = \left\| \Delta W + \eta \sum_{i=1}^N c_i \nabla f(\theta_0; x_i) \right\|^2$$

This loss has a fundamental mixing symmetry: any linear recombination of per-sample gradients that sums to the same total gradient gives the same loss value. The optimizer finds a blended local minimum where each x_i absorbs features from multiple true training images, rather than cleanly separating them. The problem is structurally identical to batch gradient inversion in federated learning, where an attacker observes the mean gradient and must recover individual images.

However, there is a critical structural advantage in the NTK regime: because the model is linearized, the weight gradient for each FC layer has a special structure. Each ROW of the weight gradient matrix is a different linear mixture of the N source images:

$$\left(\frac{\partial \mathcal{L}}{\partial W} \right)_j = \frac{1}{N} \sum_{i=1}^N \frac{\partial \mathcal{L}}{\partial y_{i,j}} \cdot x_i$$

With layer width d_{out} ($= 1000$ in our MLP), this gives d_{out} independent observations of the N -way mixture. As long as $d_{\text{out}} \geq N$, standard signal separation techniques (ICA, NMF) can analytically decompose the mixture.

9.2 Decomposition Approaches (by N)

Small N (2–10): Optimization-Based Separation

- Cross-gradient orthogonality penalty: add $\cos_{\text{sim}}(\nabla f(\theta_0; x_1), \nabla f(\theta_0; x_2))$ to

the loss. Forces images to produce orthogonal gradients, directly attacking the superposition mechanism. Most theoretically principled.

- Label-based grouping: in binary classification, $c_i > 0$ for class 0 and $c_i < 0$ for class 1. Separate positive/negative gradient contributions first, then decompose within each class. Halves the effective problem size for free.
- Diversity penalty: existing `get_diversity_penalty()` in `ntk_extraction.py` is implemented but NOT wired into the extraction loop. Lowest-hanging fruit.

Medium N (5-20): Algorithmic Decomposition

- Sequential peeling with joint refinement: reconstruct images one at a time from the residual (like matching pursuit), then jointly optimize all N using the greedy solutions as warm start. Each sub-problem is $N=1$ where the pipeline achieves $SSIM \sim 0.83$.
- Alternating optimization: fix all images except x_i , optimize x_i against the residual. Cycle through all images. Each step is a single-image problem.
- Overcomplete slots + clustering: optimize for $N'=2N$ image slots, then cluster similar results by SSIM. Extra slots absorb garbage; redundancy helps coverage.

Large N (10-1000): Analytical Separation (ICA)

- Cocktail Party Attack (Kariyappa et al., ICML 2023): apply FastICA to the FC layer weight gradient matrix. Each row is a linear mixture of N sources. With width=1000, separates up to $N=1000$. See Section 9.3.
- SPEAR (NeurIPS 2024): SVD of the gradient matrix + ReLU sparsity filtering. Each training input activates a unique subset of neurons, creating a "fingerprint." Achieves EXACT recovery for $N \leq 25$ on FC+ReLU networks.
- ARES (2025): sparse recovery in DCT basis. Scales to $N=384$ on two-layer networks.

9.3 The Cocktail Party Attack (ICA)

The Cocktail Party Attack (CPA) is the most directly applicable technique for general N. The key mathematical insight: for a fully connected layer with d_{out} output neurons, each output neuron j provides one observation of the N-way mixture:

$$(\nabla_W \mathcal{L})_j = \frac{1}{N} \sum_{i=1}^N \frac{\partial \mathcal{L}}{\partial y_{i,j}} \cdot x_i \quad (d_{out} \text{ observations of } N \text{ sources})$$

This is exactly the Independent Component Analysis (ICA) setup: d_{out} observations of N sources with unknown mixing coefficients. FastICA recovers the sources (individual images x_i) up to permutation and scaling, as long as

$d_{\text{out}} \geq N$ and the sources are statistically independent.

Why this works especially well for our setting:

- Our MLP has width 1000 $\rightarrow$ 1000 observations for separating N sources
- MNIST digits are sparse (mostly black pixels) and statistically independent
- NMF (Non-Negative Matrix Factorization) can replace ICA for non-negative data like images
- For ViT: attention layers have $d_{\text{out}} = 768$ or 3072, supporting N up to hundreds

CPA for our NTK attack — implementation sketch:

1. Run standard NTK extraction (joint optimization) to get approximate ΔW
2. Reshape ΔW for the widest FC layer as a $(d_{\text{out}} \times d_{\text{in}})$ matrix
3. Apply FastICA($n_{\text{components}}=N$) or NMF($n_{\text{components}}=N$) to the rows
4. Each recovered component is one training image (up to permutation)

CPA was demonstrated at batch size $N=1024$ in the original paper. The quality degrades as N approaches d_{out} (the system becomes barely determined), but for realistic few-shot scenarios ($N=5-50$), the system is heavily overdetermined.

9.4 Phase Transition & Scaling Limits

The practical limits of batch reconstruction depend on the method and architecture:

Method	Type	Max N	Architecture
SPEAR (2024)	SVD+sparsity	$N \leq 25$	FC + ReLU
CPA/ICA (2023)	Analytical	$N \leq 1024$	FC layer req.
GradInversion (2021)	Optimization	$N \leq 48$	ResNet+BN
ARES (2025)	Sparse recov.	$N \leq 384$	FC / CNN
Our NTK (current)	Optimization	$N=1-2$	MLP (width 1000)

Theoretical phase transition:

The fundamental limit is $N < \text{network width } (d_{\text{out}})$. For our 784-1000-1000-1 MLP, the theoretical limit is $N < 1000$. For ViT-B/16 with $d_{\text{model}}=768$, the limit is $N < 768$. Wang et al. (AISTATS 2023) proved that training data is identifiable from a single gradient query when width $m > N$, under mild conditions.

Intrinsic dimension argument (revisited):

The phase transition in raw dimension ($N < d_{\text{out}}$) is conservative. Natural images have intrinsic dimension far lower than pixel count: $\sim 10-20$ for MNIST

digits, ~ 100 – 1000 for faces. In intrinsic-dimension terms, a width-1000 layer provides ~ 50 – $100\times$ more observations than needed for $N=10$ MNIST digits. The practical bottleneck is optimization (non-convex landscape with many local minima), not information.

NEXT STEP: Wire `get_diversity_penalty()` into extraction loop, test ICA post-hoc on existing $N=2$ results

10. Critical New Literature (2023-2025)

Several recent papers (discovered April 2026) are directly relevant to the thesis and were not known during the initial planning phase. These validate the thesis direction and provide concrete techniques for scaling beyond $N=1$.

10.1 Gradient Inversion on PEFT (CVPR 2025)

Sami et al. — "Gradient Inversion Attacks on Parameter-Efficient Fine-Tuning"

The single most important validation of our thesis direction. Key findings:

- PEFT's dimensionality reduction FOCUSES gradient information, making inversion EASIER than full fine-tuning — not harder, as previously assumed
- A malicious server can design adapter architecture to maximize vulnerability
- Recovers image patches from batches up to $N=128$ on CIFAR-100
- LPIPS = 0.08 (high visual quality), 72.6% patch recovery rate

This directly contradicts the assumption that LoRA's low rank is a privacy defense. The low-rank constraint concentrates gradient signal into fewer dimensions, making each dimension more informative. Our NTK-LoRA results (SSIM 0.557 at $r=8$) are consistent with this finding.

10.2 Cocktail Party Attack (ICML 2023)

Kariyappa et al. — ICA-based gradient inversion

Demonstrates that ICA can separate individual training samples from averaged/aggregated gradients in federated learning. Each row of the FC layer gradient is a linear mixture of the N source inputs. With $d_{\text{out}} \geq N$, FastICA recovers all sources. Tested up to $N=1024$. Code available at github.com/facebookresearch/cocktail_party_attack.

10.3 SPEAR (NeurIPS 2024)

Dimitri et al. — Exact batch recovery via SVD + sparsity

First method to achieve EXACT reconstruction for batch size $N>1$. Uses SVD to find the N -dimensional subspace of training inputs in the gradient, then ReLU activation sparsity patterns as per-image "fingerprints" to filter incorrect candidates. Exact recovery for $N \leq 25$ on ImageNet inputs (FC + ReLU networks).

10.4 ARES (2025)

Scalable gradient inversion via sparse recovery

Reformulates batch inversion as noisy sparse recovery in a DCT or learned basis. Achieves PSNR ~ 90 – 120 dB at $N=32$ and 80%+ recovery at $N=384$ with two-layer networks. Currently the most scalable practical attack.

10.5 ReCIT (2025)**Reconstruct private data from PEFT gradients**

Directly relevant: reconstructs private training data specifically from PEFT/LoRA gradients. Complements our NTK approach with a different reconstruction mechanism.

10.6 Implications for the Thesis

These papers collectively validate that: (1) LoRA/PEFT is not a privacy defense — it may make attacks easier; (2) analytical decomposition (ICA, SVD) can scale $N>1$ reconstruction far beyond optimization-based approaches; (3) the NTK regime's linearity is a feature, not a limitation, because it enables ICA-style separation. The thesis should integrate the Cocktail Party Attack as the primary technique for $N>1$ scaling, with the NTK extraction providing the per-sample reconstruction and the ICA providing the decomposition.

11. Timeline & Next Steps

11.1 Immediate Actions (This Week)

- Submit Phase 0 (fixed) to WEXAC — `scripts/run_phase0_fixed_wexac.sh`
- Extend `run_experiment_b.py` for ViT support while waiting for Phase 0 results
- Read Geiping et al. gradient inversion paper (essential for Phase 0 understanding)
- Update thesis prospectus based on current results

11.2 Decision Points

Milestone	Timeline	Decision
Phase 0 result	Week 1	Fork: $SSIM > 0.3 \rightarrow$ all paths; $SSIM < 0.3 \rightarrow$ KKT only
ViT NTK $T=1$	Week 3	Go/no-go on NTK scaling
Multi-step ViT	Week 5	Proceed to Gradient Bridge?
Thesis narrative	Week 8	Finalize which directions to present

11.3 Research Backlog

- $N > 1$ decomposition: wire diversity penalty, test ICA/NMF post-hoc, implement CPA
- Characterize N vs. SSIM phase transition curve on MNIST MLP
- Image-domain priors: TV, LPIPS, SDS from frozen diffusion (critical for ViT scale)
- Read Cocktail Party Attack, SPEAR, PEFT Inversion (CVPR 2025)
- Phase transition r^* verification: sweep $r \times N$ on MLPs
- Write LaTeX summary for `notes/lora_reconstruction_writeup.tex`

11.4 What This Thesis Fills

The thesis fills a specific gap in the literature:

- Haim et al.: requires 1M+ epoch convergence (unrealistic for modern fine-tuning)
- Gradient inversion: requires actual gradient (not available from published adapter)
- This thesis: reconstructs from adapter weights — few-shot regime is most potent

No prior work has demonstrated training data reconstruction from LoRA adapter weights. The NTK approach (empirically proven on MNIST) and the Gradient Bridge (if Phase 0 succeeds on ViT) would both be novel contributions.

End of Document

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